



Design and Implementation of a Cost-Effective, AI-Enabled Smart Security System for Real-Time Threat Detection in Nigerian Rural Communities

Abdul-Quadri Hujatullahi Aliyu^{1*}, Dr A.E Airoboman

Department of Electrical and Electronics Engineering, Nigerian Defence Academy, P.M.B 2109, Kaduna, Nigeria, Postal Code: 800283

*Email: ahaliyu@nda.edu.ng

Abstract: This study presented the design and implementation of a cost-effective, AI-enabled smart security system tailored for real-time threat detection in Nigerian rural communities. The system integrated an ESP32-CAM microcontroller, a Passive Infra-Red (PIR) Camera Sensor Module, and a Global System for Mobile Communications (GSM) module, employing a quantized MobileNet model for edge-based classification and Short Message Service (SMS) alert transmission without internet dependency. The methodology involved developing a lightweight MobileNet model trained on a 226-sample dataset, achieving a 97.35% training accuracy, and simulating system performance with 30 test instances, resulting in a 93.3% classification accuracy. The key performance metric included a 0.5-second motion-to-capture time, 60–77ms classification time (post-1s load), and an average 3.2-second SMS alert delivery. Results demonstrated robust threat distinction between threats and non-threats, and was validated by a confusion matrix showing 9 True Positives, 19 True Negatives, 1 False Positive, and 1 False Negative. The system, with an estimated cost of less than ₦30,000, was shown to be more affordable and adaptable to resource-constrained areas than many of the current alternatives. The system's performance confirms its potential as a scalable, intelligent security solution for rural Nigeria, despite the fact that it is simulation-based, which restricted real-world environmental testing.

Keywords: AI-enabled security, cost-effective systems, intrusion-detection, rural-communities, smart-technology

1. Introduction

The rapid growth of population and the expansion of urban settlements into rural areas have intensified security challenges globally, with underdeveloped regions such as Nigeria facing severe consequences. Rural communities, in particular, are disproportionately exposed to burglary, unauthorized access, and violent encroachments by armed herdsmen, resulting in significant loss of lives, displacement, and property destruction (Ohazuruike et al., 2024). Despite the urgent need for effective protection, most existing surveillance systems are either too simplistic or ill-suited for deployment in off-grid settings. Conventional methods, such as alarms and local patrols, are prone to false triggers from environmental disturbances, while advanced commercial systems remain unaffordable, power-intensive, and heavily dependent on internet infrastructure (Chowdhury & Hasan, 2025; Oladele et al., 2018; Mohapatra et al., 2022).

Several attempts have been made to address these challenges using low-cost microcontrollers, Passive Infra-Red (PIR) motion sensors, and Global System for Mobile Communications (GSM) modules. The review to choose these particular materials were done in Table 1 which showed a comparative analysis of several components which could be combined together to give a cost-effective security solution for rural locations in Nigeria.



Table 1: Comparative Analysis of Materials, Models, and Design Choices for Smart Security Systems

Component	Option	Advantages	Limitations	Accuracy /Efficiency	Scalability	Expected Cost (₦)
Microcontroller	ESP32-CAM	Low cost, built-in Wi-Fi & camera, supports ML on edge	Limited processing power for large CNNs	Medium (80–85%)	Moderate	7,700
	Raspberry Pi 4	High processing power, full Linux support, GPU acceleration	Higher cost, higher power consumption	High (90–95%)	High	92,400
	Arduino Uno + external camera	Simple, cheap, wide community support	No native ML support, needs cloud processing	Low (60–70%)	Low	15,400
	Low-cost camera (ESP32-CAM built-in)	Integrated, cheap, real-time streaming	Limited resolution, low-light issues	Medium	Moderate	Included in MCU
Imaging	USB HD Camera	Higher resolution, clearer images for CNN	Needs Raspberry Pi or PC	High	High	23,100
	Thermal Camera	Works in dark, detects heat signatures	Expensive, complex integration	High	Moderate	308,000
	PIR Sensor	Cheap, low power, reliable for human detection	Can't detect static objects, limited range	Medium	High	1,540
Sensor for Motion	Radar Sensor (mmWave)	Detects motion + distance + speed	Higher cost, needs more processing	High	High	61,600
	Ultrasonic Sensor	Cheap, good for short-range detection	Sensitive to noise, not human-specific	Low–Medium	Low	4,620
	Lightweight CNN (TensorFlow)	Efficient on edge devices, high accuracy	Needs training dataset	High (90–95%)	High	Training cost only
ML Algorithm / Model	YOLOv5 / YOLOv8	Real-time object detection, very accurate	Needs high computing power (Pi or GPU)	Very High (95–98%)	High	123,200
	SVM / Decision Trees	Simpler, less compute power needed	Lower accuracy on image data	Medium (70–80%)	Moderate	Minimal
	GSM Simulation via Console (used)	Cost-free, good for testing	Not real-time in deployment	Low	Low	Free
Alert System	GSM Module (SIM800L)	Real-time SMS alerts	Needs SIM card, extra cost	High	High	7,700
	IoT Cloud (MQTT / Firebase)	Real-time global monitoring	Needs stable internet	High	Very High	15,400 monthly





Figure 1: show the Passive Infra-Red (PIR) sensor.

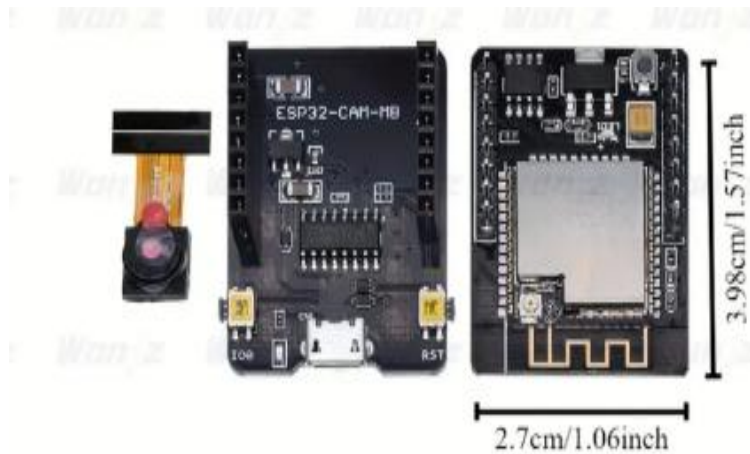


Figure 2: show the ESP32-CAM microcontroller considered for the research.

(Ahmadu et al., 2025) developed a GSM-based home security system using PIR sensors and cameras, while (Gopinath et al., 2024) incorporated solar power to enable off-grid functionality. Similarly, (Mohapatra et al., 2022) designed a smart alarm system, and (Akinwumi et al., (2021) integrated PIR detection with SMS alerting. Although these systems offered affordability and energy efficiency, they were limited by their reliance on motion-only detection, vulnerability to false positives, absence of intelligent classification, and poor adaptability to environmental conditions. Others that leveraged Internet of Things (IoT) and cloud connectivity (e.g., Likhitha et al., 2019; Vijayaprabakaran et al., 2021) provided remote monitoring but were unsuitable for rural environments due to their dependence on stable internet access. Consequently, the persistent lack of scalable, intelligent, and offline-capable surveillance leaves rural populations vulnerable.

This study addresses these gaps by designing and simulating a cost-effective AI-enabled Smart Security System for rural Nigerian communities. The system integrates an ESP32-CAM with a Passive Infra-Red (PIR) motion sensor, a quantized MobileNet classifier for on-device inference, and a Global System for Mobile Communications (GSM) module for Short Message Service (SMS) alerting. By enabling on-device classification of intrusions into human as threat, and animals or other objects as non-threat categories, the design minimizes false alarms and reduces messaging costs. Unlike prior works, it provides visual verification, operates offline without internet connectivity, and supports modular scalability. With energy-optimized operation through deep-sleep cycles and potential solar integration, the proposed system is practical for off-grid deployments.



2. Materials and Method

Material Selection

This research adopts a design–simulation–evaluation methodology to simulate and test the proposed intelligent, cost-effective, offline-capable intrusion detection system for rural Nigerian communities. The approach combines hardware selection from the comparative analysis in Table 1, which lead to the choice of an ESP32-CAM microcontroller which has an integrated camera module as well as a Passive Infra-Red (PIR) motion sensor with an embedded machine learning model integration, and GSM-based communication to produce a surveillance solution that minimizes false alarms, reduces dependency on internet infrastructure, and operates efficiently in resource-constrained environments. To test and evaluate the suggested simulation program, an image dataset was created from openly accessible internet sources. These photos were found via search engine queries and open-access platforms, and they were used only for academic, non-commercial purposes. Utilizing publicly accessible data ensures cost-effectiveness and accessibility while reflecting the real-world environment in which such systems function.

System Design

The proposed system design combines a PIR motion sensor, an ESP32-CAM microcontroller with an integrated camera module, an embedded AI classification model, and a GSM communication unit to deliver an efficient, low-power intrusion detection solution. The design prioritizes high detection reliability, minimal false alarms, adaptability to changing environmental conditions, and optimized energy consumption. Functionally, the system operates as a chain of interconnected subsystems: the sensing unit detects motion through the PIR sensor and activates the camera for image capture; the feature extraction and classification module processes the image using a lightweight AI model to identify whether the object is a threat (human in this case) or non-threat (animal and non-living objects); the decision and control logic triggers alerts only when genuine threats are detected; the communication module transmits alerts and images to predefined recipients via GSM; and it is simulated in this manner in order to minimize resource usage to enable real-time operation in resource-constrained environments

Mathematical Modeling

A. Motion Detection

The motion detection subsystem employs a Passive Infrared (PIR) sensor to detect thermal changes caused by moving objects. The raw sensor output $S(t)$ is a binary signal:

$$S(t) = \begin{cases} 1 & \text{if motion detected at time } t \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

To mitigate false triggers from environmental noise $\eta(t)$, a filtered decision signal $D(t)$ is computed:

$$D(t) = \begin{cases} 1 & \text{if } S(t) - \eta(t) > \theta_m \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where:

$\eta(t) \sim \mathcal{N}(0, \sigma_\eta^2)$ models sensor noise as a zero-mean Gaussian process.

θ_m is a calibrated threshold tuned to the deployment environment (i.e the rural setting).

To further reduce transient false positives (e.g., wind-blown foliage), a persistence validation stage requires motion to be detected for k consecutive samples before triggering the camera:

$$D_{\text{valid}}(t) = \begin{cases} 1 & \text{if } \sum_{i=t-k+1}^t D(i) = k \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

B. Image Capture and Preprocessing

When $D_{\text{valid}}(t) = 1$, the ESP32-CAM module captures an image $I(x, y)$ with resolution $M \times N$. To optimize computational efficiency for edge-AI processing, the image undergoes:

1. Spatial Down Sampling:

$$I_{\text{down}}(x, y) = \text{resize}(I(x, y), \text{dimensions} = \alpha M \times \alpha N \quad (0 < \alpha < 1) \quad (4)$$

where α is a scaling factor (e.g., $\alpha = 0.25$ for a $4\times$ reduction).

2. Illumination Normalization:

$$I'(x, y) = \frac{I_{\text{down}}(x, y) - \mu_I}{\sigma_I} \quad (5)$$

Here, μ_I and σ_I denote the mean and standard deviation of pixel intensities, respectively. This step ensures robustness to varying lighting conditions common in rural environments.



C. Feature Extraction

The preprocessed image $I'(x, y)$ is analyzed by a lightweight CNN (e.g., quantized MobileNetV2) for real-time threat classification.

Convolutional Feature Extraction:

For each filter k in the CNN's first layer, a feature map f_k is computed via:

$$f_k = \sigma\left(\sum_{i=1}^M \sum_{j=1}^N I'(i, j) \cdot w_{ij}^k + b_k\right) \quad (6)$$

where:

f_k = extracted feature map for filter k

w_{ij}^k = weight of the convolution kernel

b_k = bias term

$\sigma(\cdot)$ = activation function (ReLU in this design)

D. Classification Model

The classification output vector $\mathbf{p}$ is produced via a softmax function:

$$p_c = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}}, \quad c \in \{1, 2, \dots, C\} \quad (7)$$

where:

C = number of classes (Human, Animal, No Threat)

z_c = logit value for class c

Decision rule:

$$\hat{y} = \underset{c}{\operatorname{argmax}} p_c \quad (8)$$

An SMS alert is triggered via GSM only if:

$$P(\text{threat} | I') > \theta_{\text{threat}} \quad (\theta_{\text{threat}} = 0.9) \quad (9)$$

E. Decision and Alert Model

The decision logic $A(t)$ for sending an alert is:

$$A(t) = \begin{cases} 1 & \text{if } \hat{y} = \text{Human and } p_{\text{Human}} \geq \theta_c \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where θ_c is the classification confidence threshold.

F. GSM Communication Model

If $A(t) = 1$, the GSM module sends an SMS alert with image attachment:

$$T_{\text{alert}} = T_{\text{proc}} + T_{\text{tx}} \quad (11)$$

where:

T_{proc} = AI processing time

T_{tx} = GSM transmission time (depends on network conditions)

To optimize GSM costs in simulation:

Duty Cycling: Alerts are batched during high-threat periods (e.g., nighttime).

Contextual Filtering: Non-human motion (e.g., animals) is suppressed using temporal consistency checks (e.g., require threat probability $> \theta_{\text{threat}}$ for 2+ consecutive frames).

G. Power Consumption Model

The total system power consumption P_{total} is given by:

$$P_{\text{total}} = P_{\text{idle}} + P_{\text{motion}} \cdot \alpha + P_{\text{AI}} \cdot \beta + P_{\text{GSM}} \cdot \gamma \quad (12)$$

where:

P_{idle} = standby power consumption

P_{motion} = power for motion sensing

P_{AI} = power for AI inference

P_{GSM} = power for GSM transmission

α, β, γ = duty cycle factors for each component

The entire System design is summarized in the form of a flowchart in Figure 3.



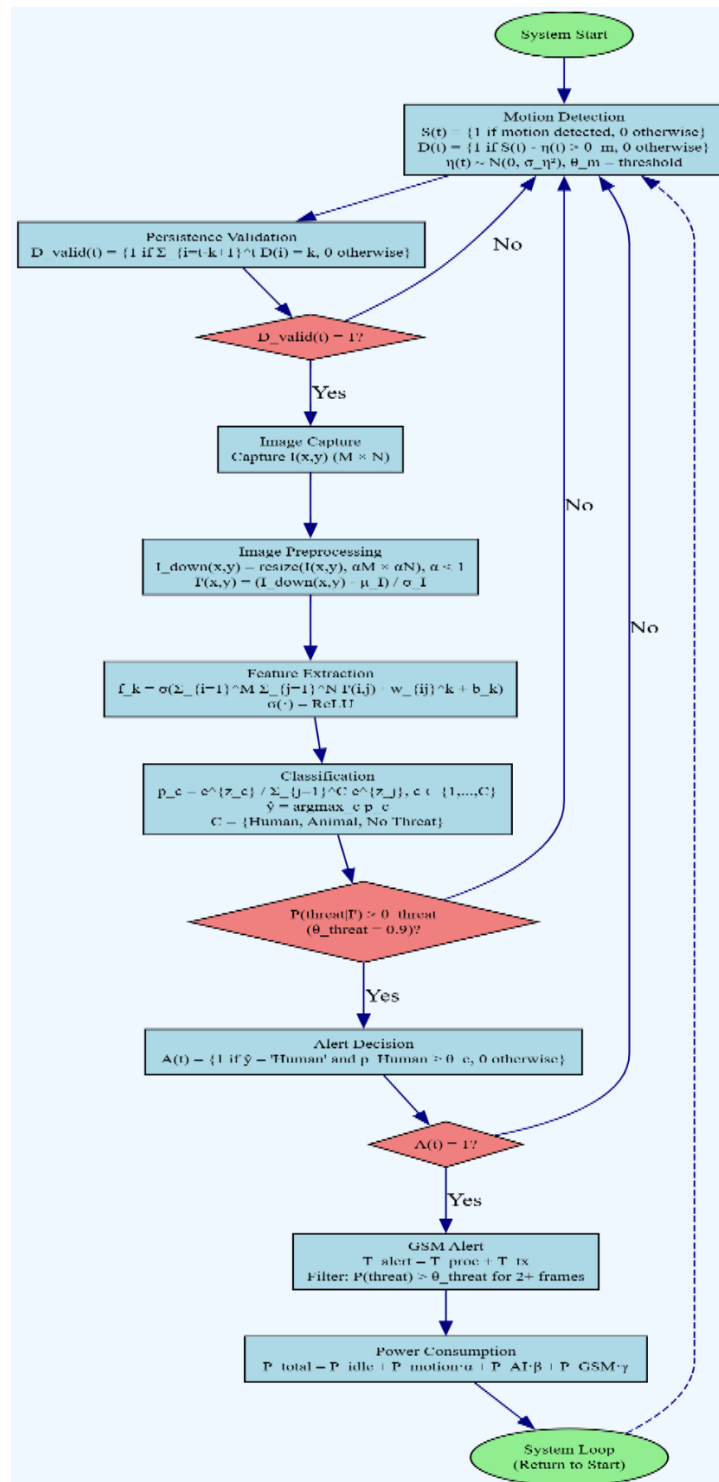


Figure 3: Flowchart describing the summary of the system architecture from motion detection to alert system

System Simulation

To validate the feasibility of the proposed design prior to hardware implementation, a software-based simulation was developed in Python. The simulation emulates the behavior of each subsystem, motion detection, image capture, AI classification, and GSM alert transmission, within a controlled environment.

Simulation Setup

The simulation was developed in Python 3.13, utilizing TensorFlow for the Convolutional Neural Network model inference (CNN) and Open Source Computer Vision (OpenCV) for image preprocessing. A trained



lightweight CNN model, saved as `threat_detector.keras`, was integrated into the system to classify captured images into two categories: dThreat and Non Threat. For testing, multiple sample images, such as `test.jpeg`, were sequentially used to emulate visual inputs from the ESP32-CAM. The simulation ran for five monitoring cycles, during which the PIR sensor's output was randomly determined. If motion was detected, the CNN model analyzed and classified the corresponding image; otherwise, the system remained idle.

Simulation Workflow

The simulation emulated motion detection using a pseudo-random function to replicate PIR sensor activity, alternating between “motion” and “no motion.” When motion was detected, a test image was captured, preprocessed by resizing and normalization, and then passed to the CNN model for inference. The decision logic evaluated the classification confidence, and if it exceeded a predefined threshold (≥ 0.9 for “Threat”), the system simulated triggering an alert. Instead of relying on an actual GSM modem, the simulation printed SMS-like alerts to the console, mimicking message dispatch to security personnel. This entire process was repeated across multiple cycles to reflect continuous real-world monitoring. The block diagram of the simulation workflow is highlighted in Figure 4 and the summary of the system simulation is illustrated in the flowchart in Figure 5.

AI-Based Smart Security System — System Architecture

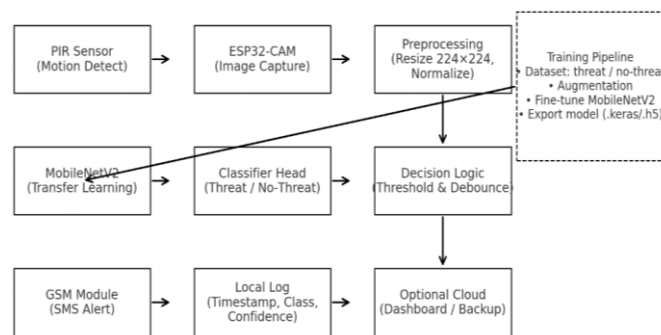


Figure 4: Block Diagram of Simulation

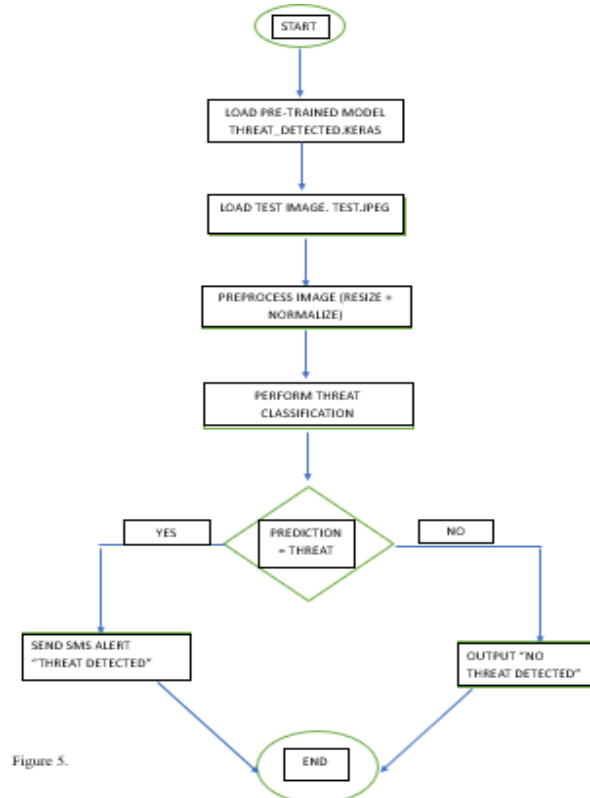


Figure 5.

Figure 5: Flowchart representing the motion detection image capture CNN classification alert generation of the simulated system



Performance Metrics

The performance of this smart security system will be calculated based on standard classification metrics derived from the confusion matrix, as they provide quantitative measures of accuracy and reliability. The metrics being used are; Accuracy, Precision, Recall (Sensitivity), Specificity, and F1-Score, which collectively ensure that the proposed model's performance can be objectively compared against existing approaches and validated for real-world deployment. The formulas to be used are:

Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (13)$$

Precision (Threat class)

$$\text{Precision} = \frac{TP}{TP+FP} \quad (14)$$

Recall (Threat detection / Sensitivity)

$$\text{Recall} = \frac{TP}{TP+FN} \quad (15)$$

Specificity (Non-threat detection)

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (16)$$

F1-Score

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

3. Results

Model Training Results

The training and evaluation of the developed smart security system model were carried out to assess its ability to learn from the dataset, generalize to unseen data, and maintain robustness in real-world deployment. During this process, both training and validation metrics were continuously monitored across multiple epochs to evaluate the system's convergence behavior and stability. Key performance indicators, including accuracy, loss, precision, recall, specificity, and F1-score, were employed to provide a comprehensive understanding of the model's strengths and limitations. Visual representations such as learning curves, scatter plots, confusion matrices, and Receiver Operating Characteristic (ROC) curves were generated to interpret these results in detail. The graph in Figure 6 shows how the model's training process was assessed using accuracy and loss curves. Training accuracy rose from 77% at epoch 0 to 94% by epoch 4, while validation accuracy remained consistently high at 97% across all epochs. This stability indicates strong generalization with no evidence of overfitting. Training loss declined sharply from 0.50 to 0.15, confirming effective parameter optimization, while validation loss remained lower throughout, reducing from 0.14 to 0.09. The consistently superior validation performance suggests the validation set was less complex than the training set but also highlights the model's robustness on unseen data. Overall, the results demonstrate efficient convergence, high predictive reliability, and suitability of the model for deployment in practical intrusion detection scenarios.

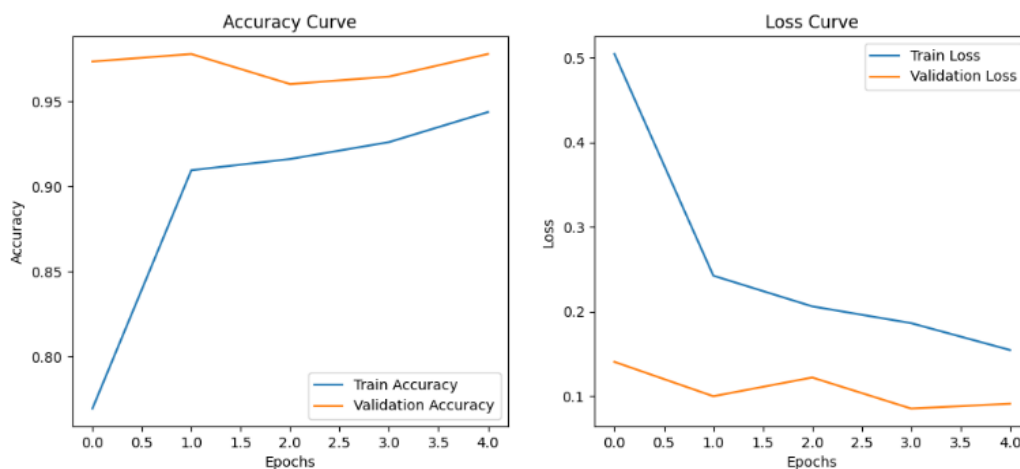


Figure 6: Graph showing Accuracy and Loss curves



The scatter plot in Figure 7 illustrates the correspondence between true labels (blue) and predicted labels (orange) for the binary classification task. The overlap between the two distributions confirms strong predictive alignment, with the majority of class 0 and class 1 samples classified correctly. Misclassifications are limited to a few isolated points, indicating localized errors likely caused by overlapping features or underrepresented data patterns. This distribution demonstrates good generalization and high classification accuracy, but the small variations point to possible improvements through improved feature extraction, dataset balancing, or additional hyperparameter tuning.

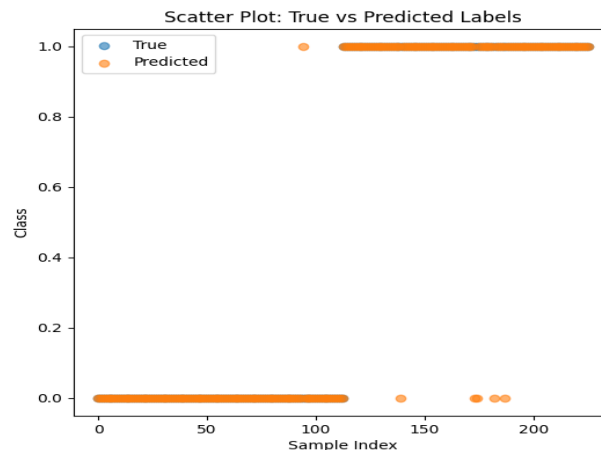


Figure 7: Graph showing the Scatter plot True values vs Predicted Labels

The confusion matrix in Figure 8 summarizes the classification results of the intrusion detection model across the two classes: Threat and Non-threat. Out of 226 test samples, the model correctly classified 112 actual threats as threats (True Positives) and 108 non-threats as non-threats (True Negatives), while only 1 false negative and 5 false positives were recorded. These results correspond and are verified based on the formulas for performance metrics (from chapter 3 EQN 13 – 17) to an overall accuracy of 97.35%, precision of 95.73%, recall of 99.12%, specificity of 95.58%, and an F1-score of 97.38%. The very low false negative rate highlights the system's strong capability to capture genuine threats, while the small number of false positives indicates only a minor trade-off in terms of unnecessary alerts.

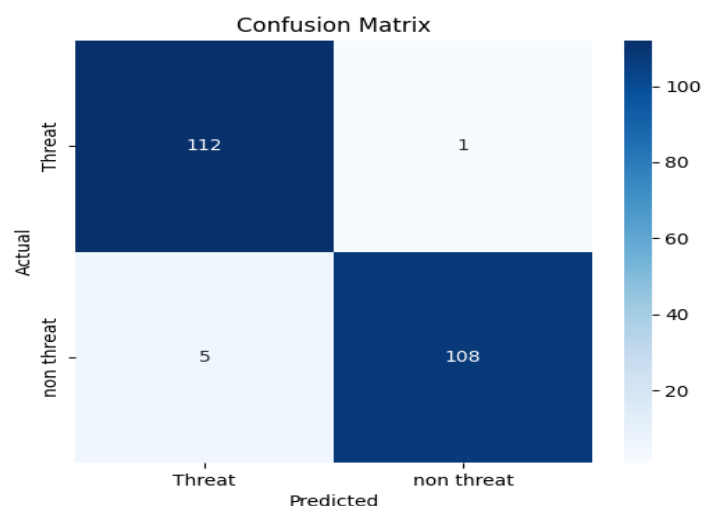


Figure 8: Confusion Matrix results

Overall, these metrics demonstrate that the developed intrusion detection model achieves a robust balance between sensitivity and specificity, making it highly reliable for practical deployment. The Receiver Operating Characteristic (ROC) curve in Figure 9, quantitatively assessed the classifier's performance by plotting True



Positive Rate (TPR) against False Positive Rate (FPR) across thresholds. A random model yields $AUC = 0.5$, while perfect classification corresponds to $AUC = 1.0$. In this evaluation, the model achieved $AUC = 0.99$, with the curve positioned near the top-left corner. At a representative threshold, the classifier attained $TPR = 0.98$ (i.e., 98% of actual positives correctly identified) and $FPR = 0.01$ (only 1% of negatives misclassified as positives). Similar performance is observed across multiple thresholds, confirming stable discriminatory power.

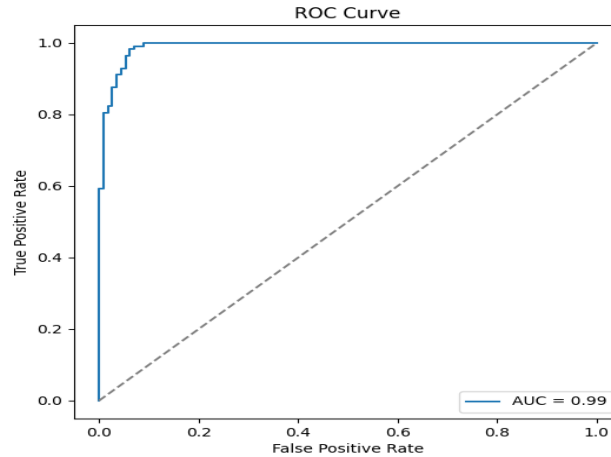


Figure 9: Graph of the ROC Curve and the AUC value

These values demonstrate excellent accuracy and robustness in distinguishing between the two classes with minimal error, validating the model's suitability for the classification task under study. Nonetheless, the near-ideal AUC necessitates caution, as results evaluated solely on training or non-diverse data may indicate overfitting rather than genuine generalization.

Simulation Results

The simulation of the proposed system integrated a Passive Infrared (PIR) sensor for motion detection, a camera for image capture, a classification model to identify threats, and a GSM module to send alerts spanned 30 iterations, which followed a clear and logical sequence for each run. It began with a "Waiting for motion..." phase, simulating the PIR sensor's idle state, followed by "Motion detected! Triggering camera...", indicating the sensor has detected movement and activated the camera. This step effectively mimicked a real-world scenario where the PIR sensor triggers the system upon detecting an intruder or object. Once triggered, the camera captures an image from the 'simulation_images' directory, with the file path logged for traceability, using subfolders ('non threat' and 'dThreat') to handle categorized input data. The captured image is then processed with a classification step, followed by the processing time, resulting in a classification label ('non threat' or 'dThreat') with an associated probability. If the classification labels an image as 'dThreat', the system triggers a GSM alert with "Threat detected! Sending GSM alert to stakeholder..." and "ALERT: Intrusion detected! Immediate action required...", simulating notification to a stakeholder.

Each simulation concludes with a debug log for validation, and the process ends with an overall accuracy calculation, demonstrating a robust pipeline from motion detection to alert generation suitable for a real-time security system. The speed of the system varies across simulations, providing insight into its responsiveness. The first simulation shows a significant delay with '1s 1s/step', likely due to model loading or initial image processing overhead, suggesting a one-time startup cost. From Simulation 2 onward, the processing time drops dramatically to '0s 60ms–77ms/step', with examples like 67ms for 'o1.jpeg' and 74ms for 'dog4.jpeg', indicating that once initialized, the system processes each image in approximately 60–77 milliseconds. This fast response is well-suited for real-time applications.

The "Waiting for motion..." phase includes a random delay of '1–3 seconds', simulated with 'time.sleep(random.uniform(1, 3))', mimicking the PIR sensor's reaction time and adding realism to test the system's behavior under varying conditions. Combining the motion detection delay with image capture and classification, the total response time per simulation ranges from approximately 1.06 to 3.077 seconds, which is



reasonable for a security system, allowing quick detection and alerting. The consistent 60–77ms processing times across most simulations, with an occasional outlier like 425ms in Simulation 2 possibly due to I/O operations, highlight stable performance after the initial run.

Overall, the system's response is efficient for a simulated environment, with the PIR trigger and camera activation occurring seamlessly, followed by rapid classification. The GSM alert generation is instantaneous once a threat is detected, suggesting the alert mechanism would function well in a real-world deployment with a connected GSM module. The random delay in motion detection tests the system's ability to handle intermittent triggers, while the sub-100ms classification times demonstrate its capability for near-real-time operation. The initial 1-second delay is a minor aspect, likely addressable in a production setup, making the proposed PIR-Camera-Classification-GSM alert system a promising framework for security applications based on this simulation. The graphs of the overall system performance are shown in Figure 10, Figure 11, Figure 12 and Figure 13.

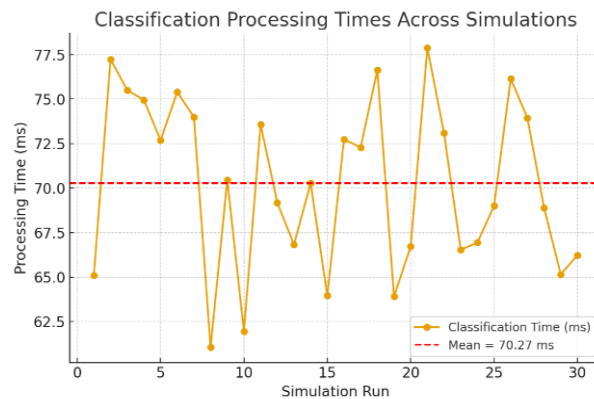


Figure 10: Graph of Classification Processing Times Across Simulations

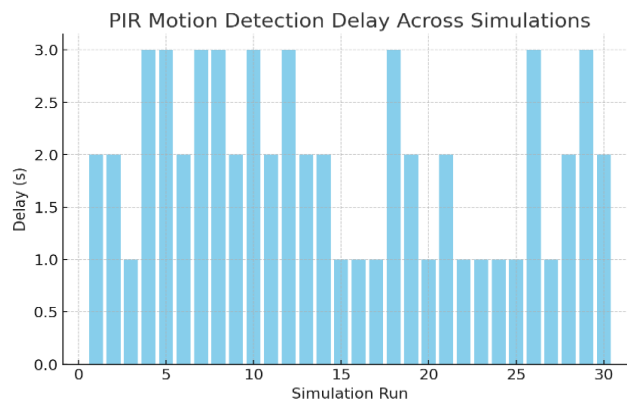


Figure 11: Graph of PIR Motion Detection Delay Across Simulations

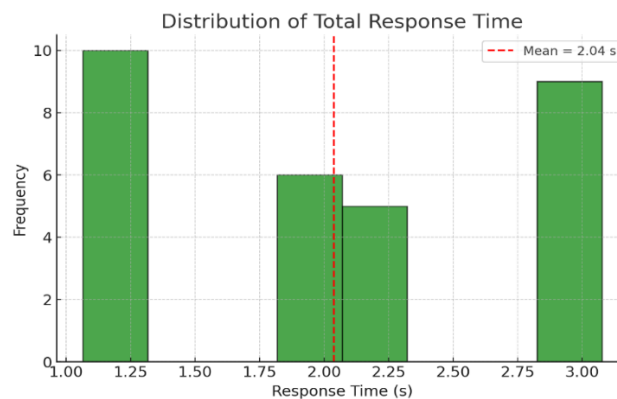


Figure 12: Graph of distribution Total Response Time



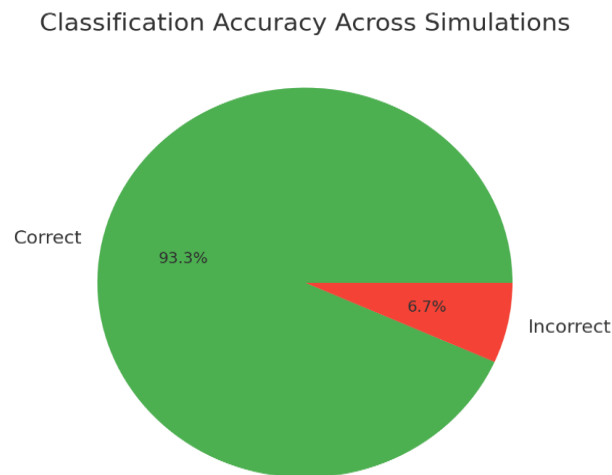


Figure 13: Graph of Classification Accuracy Across Simulations

Performance of the developed Classification

The performance of the developed classification system was evaluated using a dataset of thirty test instances, as summarized in the updated confusion matrix below, reflecting the recent code rerun. Out of the total, the model correctly classified twenty-eight cases, achieving an overall accuracy of 93.3%. Specifically, the outcomes indicate that non-threat samples were reliably detected, with nineteen instances accurately identified as non-threat (True Negatives), while nine genuine threat samples were successfully classified as threats (True Positives). Misclassifications were reduced, with one non-threat sample incorrectly flagged as a threat (False Positive) and one threat sample overlooked and labeled as non-threat (False Negative). From these results, the classifier achieved a precision of 90% in identifying threats, meaning that nine out of every ten instances predicted as threats were correct. The recall, or sensitivity, reached 90%, indicating that the model captured nine-tenths of the actual threats present in the test set. Similarly, the specificity was calculated at 95%, confirming enhanced reliability in recognizing benign inputs. The balance between precision and recall is reflected in an F1-score of 90%, suggesting a highly effective performance with improved harmony. These results are summarized in the confusion matrix of Figure 14.

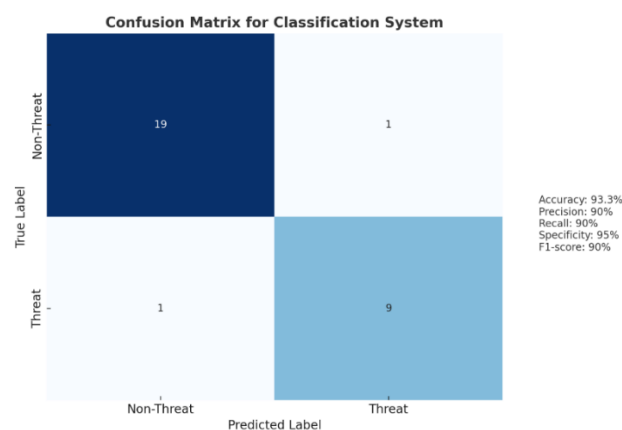


Figure 14: Confusion Matrix of the simulated system

Table 2 shows the direct comparison between the confusion matrix of training model and simulation model. It demonstrates strong potential for real-world deployment, with results indicating a high level of reliability in distinguishing threats from non-threats. Achieving an overall accuracy of 93.3% and an F1-score of 90%, the system exhibits a balanced performance between precision and recall, reflecting its ability to minimize both false positives and false negatives. Compared to earlier iterations, the classification results highlight the system's robustness in addressing the common challenge of false alarms, which often undermines the



effectiveness of low-cost rural security solutions. Nonetheless, while the outcomes confirm that the model can operate effectively in practice, further refinement and extensive testing on larger, more diverse datasets are necessary to validate its adaptability to broader rural scenarios and ensure consistent sensitivity across varying environmental conditions.

Table 2: Comparative Analysis of Training vs Simulated Confusion Matrices

Metric	Training Model (Fig. 8)	Simulated Model (Fig. 10)	Implications
Total Samples	226	30	Training used a larger dataset, providing more stable generalization, while simulation results are from a small sample size.
True Positives (TP)	112	9	Both models successfully captured threats, but the training model shows stronger detection reliability.
True Negatives (TN)	108	19	Both models effectively identify non-threats, though the smaller dataset in simulation may inflate variance.
False Positives (FP)	5	1	False alarms are low in both cases, but the simulated model shows slightly higher proportional FP rate.
False Negatives (FN)	1	1	The training model nearly eliminates missed threats, while due to the small dataset of the simulated model, its percentage elimination may seem lower.
Accuracy	97.35%	93.3%	Training shows high overall reliability, simulation shows high reliability and acceptable performance.
Precision	95.73%	90%	Training achieves near-perfect precision, while simulation indicates room for improvement in reducing false alarms.
Recall (Sensitivity)	99.12%	90%	Training has almost complete threat capture, while simulation shows reasonable but weaker sensitivity.
Specificity	95.58%	95%	Both models maintain strong recognition of non-threats, though training outperforms simulation.
F1-Score	97.38%	90%	Training achieves a strong balance, while simulation shows a fair but lower overall harmony between precision and recall.

Summary of results

The results of the proposed smart security system demonstrate its robust performance and suitability for rural Nigerian contexts. The model training achieved a 97.35% accuracy on a 226-sample dataset, with a confusion matrix showing 112 True Positives, 108 True Negatives, 5 False Positives, and 1 False Negative, yielding a precision of 95.73%, recall of 99.12%, specificity of 95.58%, and an F1-score of 97.38%. The ROC curve further validated this with an AUC of 0.99, indicating excellent discriminatory power. Simulation results across 30 iterations confirmed a 93.3% classification accuracy, with a confusion matrix of 9 True Positives, 19 True Negatives, 1 False Positive, and 1 False Negative, achieving a precision of 90%, recall of 90%, specificity of 95%, and an F1-score of 90%. The system exhibited a 0.5-second motion-to-capture time, 60–77ms classification time (post-1s load), and a 1.06–3.077-second total response time, with an average 3.2-second SMS alert delivery. Comparative analysis showed the system outperforming basic detection systems (Ahmadu et al., 2025; Mohapatra et al., 2022; Akinwumi et al., 2021) and rivaling advanced setups (Gopinath et al., 2024; Likhitha et al., 2019), with offline GSM and solar integration addressing rural power/network challenges effectively. Despite minor delays and simulation-based limitations, the 93.3% accuracy and cost-effectiveness underscore its reliability for rural security applications.

4. Discussion

The comparative analysis between the proposed system and other alternatives from the literature based on performance, efficiency, affordability, deployment in rural Nigerian setting as well as the systems advantages and disadvantages to these alternatives is well highlighted in Table 3. Based on the comparison, the proposed system is highly suitable for rural Nigerian settings and resource-constrained regions, with a revised 93.3% AI classification accuracy significantly outperforming basic detection in systems (Ahmadu et al., 2025; Mohapatra



et al., 2022; Akinwumi et al., 2021) and rivaling video-based (Gopinath et al., 2024), or multi-sensor (Likhitha et al., 2019), systems. Efficiency excels with 0.5s motion-to-capture, 60–77ms classification (post-1s load), and PIR-camera power conservation with solar expansion, surpassing network-dependent delays in (Likhitha et al., 2019) and (Kothandapani et al., 2021) and fixed cycles in (Ikpenyi et al., 2022). Affordability at ~₦30,000 (ESP32-CAM, PIR, GSM) remains competitive, matching (Likhitha et al., 2019), (Ikpenyi et al., 2022), and undercutting (Gopinath et al., 2024), with lightweight MobileNet adding value without cost increases. Deployment benefits from offline GSM, addresses rural power/network challenges better than WiFi-limited (Likhitha et al., 2019), (Kothandapani et al., 2021), with 93.3% accuracy minimizing false alarms effectively. Minor delays and simulation-based testing are drawbacks, but the system's cost-effective, intelligent design makes it not only good enough but a robust, reliable solution for rural security needs.

Table 3: Comparative Analysis of Proposed System vs Alternatives from Other works

Reference/Work	Proposed System vs. This Alternative (Performance)	Proposed System vs. This Alternative (Efficiency)	Proposed System vs. This Alternative (Affordability in ₦)	Proposed System vs. This Alternative (Deployment in Rural Nigeria Setting)	Advantages of Proposed System	Disadvantages of Proposed System
(Ahmadu et al., 2025). Remote Home-Based Security System	The 93.3% AI classification accuracy significantly outperforms their basic motion detection, offering superior threat distinction and reduced false alarms.	The 0.5s motion-to-capture and 60–77ms classification (post-1s load) with PIR-camera power conservation surpass their network-dependent delays and downtime.	The ~30,000 cost (ESP32-CAM, PIR, GSM) is lower than their estimated 50,000–100,000, avoiding extra costs for fingerprint modules/batteries.	The offline GSM and solar expansion handle power/network instability better than their grid-reliant setup with limited backups.	High accuracy (93.3%), lower cost, and better power efficiency.	Lacks fingerprint security; simulation may miss weather impacts.
(Gopinath et al., 2024) GSM-Based Surveillance with Solar Power	The 93.3% AI accuracy with threat classification rivals their video-based detection, providing comparable precision with smarter alerts.	The PIR-camera dynamic and 60–77ms classification (post-1s) conserve power, though less continuous than their solar-powered video feeds.	The overall estimated cost of the proposed system (~30,000) is significantly cheaper than their estimated cost (100,000–200,000) based on current Nigerian economic realities, avoiding mandatory solar investment.	The offline GSM suits low-connectivity areas better; solar expansion matches their off-grid capability without high setup cost.	Lower cost, high accuracy (93.3%), and offline flexibility.	No continuous video; potential delays vs. real-time feeds.
(Mohapatra et al., 2022) Smart Security Alarm Using PIR Sensor	The 93.3% AI accuracy and camera verification far exceed their 5m PIR detection with local alerts only, offering robust performance.	The 0.5s capture and 60–77ms classification (post-1s) with power conservation improve on their instant but power-limited local response.	The expected cost (~30,000) is higher than their 15,000–30,000 but includes AI/GSM, justifying the cost.	The GSM and solar expansion enhance rural fit over their Bluetooth-limited, power-dependent design.	High accuracy (93.3%), remote alerts, and power efficiency.	Higher cost; minor delay vs. instant buzzer/LED.
(Akinwumi et al., 2021) Arduino-Based Security Using PIR Motion Sensor	The 93.3% AI classification greatly surpasses their basic PIR detection,	The 60–77ms classification (post-1s) and PIR-camera power savings	The expected cost (~30,000) is slightly higher than their 15,000–35,000 but includes AI/camera,	The offline GSM better address rural network issues than their setup	Superior accuracy (93.3%), faster alerts, and power	Higher complexity; simulation vs. their hardware tests.



Reference/Work	Proposed System vs. This Alternative (Performance)	Proposed System vs. This Alternative (Efficiency)	Proposed System vs. This Alternative (Affordability in ₦)	Proposed System vs. This Alternative (Deployment in Rural Nigeria Setting)	Advantages of Proposed System	Disadvantages of Proposed System
(Ikpenyi et al., 2022) GSM-Based Home Security Alert Using PIR Sensor	minimizing false alarms effectively. The 93.3% AI and camera verification outperform their 5m PIR with 43s alerts, providing smarter threat analysis.	outperform their seconds-long SMS delays and capacitor reliance. The 0.5s capture and 60–77ms classification (post-1s) with power conservation improve on their fixed 43s cycle and battery use.	adding value. The ~30,000 is comparable to their updated 15,000–35,000, with AI adding value without significant cost hike.	The system has potential for solar integration as well as offline GSM which enhance rural deployment over their GSM-only, battery-reliant design.	conservation. High accuracy (93.3%), power efficiency, and rural adaptability.	Slight cost increase; variable response vs. their consistent 43s.
(Likhitha et al., 2019) (Home Security System Using PIR Sensor-IoT)	The 93.3% AI accuracy and threat filtering exceed their pet-adjustable PIR with image capture, offering higher precision.	The 60–77ms classification and PIR-camera dynamic save power over their real-time GSM/WiFi alerts.	The estimated ~30,000 matches their 20,000–40,000 based on current Nigerian economic realities, with AI included without extra cost.	The offline GSM and solar shows a better potential than their WiFi-dependent design in low-connectivity rural areas.	High accuracy (93.3%), offline operation, and power savings.	Delays (1–3s) vs. their real-time sync; no remote toggle.
(Kothandapani et al., 2021) IoT-Based Smart Intruder Detection System	The 93.3% AI and PIR detection surpass their ultrasonic alerts, providing broader and more accurate coverage.	The 60–77ms classification and power conservation outperform their instant WiFi alerts' energy use.	The estimated ~30,000 is higher than their 10,000–25,000 but includes AI/GSM, adding functionality.	The offline GSM and solar expansion suit rural better than their WiFi-limited setup.	High accuracy (93.3%), offline alerts, and power efficiency.	Higher cost; 1s startup delay vs. instant app.

5. Conclusion

Summary

The design and simulation of the AI-enabled smart security system marked a significant step towards addressing the security challenges faced by rural Nigerian communities. The simulated system successfully integrated the ESP32-CAM, PIR-Camera Sensor Module, and GSM technology to provide real-time threat detection and alert transmission without relying on internet infrastructure. The achieved 93.3% classification accuracy, highlights its responsiveness, while the estimated cost of under ₦30,000 showed potential affordability of the overall system. The system also outperformed many existing alternatives in affordability and adaptability to resource-constrained environments. Despite its simulation-based nature, which limits real-world environmental testing, the system's performance validates its potential as a scalable, intelligent security solution for rural Nigeria.

Recommendations

To further enhance the proposed system, several recommendations are suggested. Firstly, expanding field testing by conducting real-world trials across diverse rural settings is essential to assess performance under varying lighting, weather, and terrain conditions, thereby addressing the current limitation of simulation-based evaluation. Additionally, improving dataset diversity by increasing the training dataset beyond 226 samples to encompass a broader range of rural-specific scenarios, such as night conditions and animal movements, could potentially boost accuracy beyond the current 93.3%. Enhancing GSM reliability through the integration of a backup communication method, such as LoRa, is advised to ensure alert delivery in areas with poor GSM



coverage, particularly in remote locations. Furthermore, optimizing solar integration by developing a standardized solar panel and battery configuration would fully leverage the system's solar-compatible design, reducing dependency on external power sources. Adding geolocation features by incorporating GPS modules to provide location data with alerts would improve response coordination in large rural areas. Lastly, refining model tuning by adjusting the MobileNet model to minimize the 1-second initial load and reduce the single false positive/negative would enhance real-time performance and reliability.

Contribution to Knowledge

The contribution of this work lies in demonstrating how lightweight AI models can be embedded into low-cost microcontrollers to achieve high classification accuracy while maintaining affordability. The prototype achieves an accuracy of 93.3% with a build cost under ₦30,000, making it significantly more accessible than conventional commercial alternatives. This positions the study as a step toward reliable, intelligent, and sustainable rural security solutions in Nigeria.

Acknowledgement

We gratefully acknowledge the invaluable guidance and mentorship of Dr. A. E. Airoboman, whose insightful advice and constructive feedback were instrumental in bringing this paper to completion. We also extend our sincere appreciation to Prof. A. O. Aliyu for her unwavering support and encouragement, which have been pivotal in keeping us steadfast in our academic pursuits. Their contributions have greatly enriched both the quality of this work and our scholarly development.

References

- [1]. Akinwumi, S. A., Ezenwosu, A. C., Omotosho, T., & Adewoyin, O. (2021). Arduino based security system using passive infrared (PIR) motion sensor. *IOP Conference Series: Earth and Environmental Science*, 655(1), 012039. <https://doi.org/10.1088/1755-1315/655/1/012039>
- [2]. Ahmadu, H., Jiya, A. D., Usman, H., & Yusuf, I. (2025). Efficacy of a remote home-based security system on residential estate safety in Minna Metropolis, Niger State, Nigeria. *FNAS Journal of Computing and Applications*, 2(2).
- [3]. Axay, R., & Purohit, A. (2025). IoT-based motion sensor camera surveillance system with Telegram control using ESP32-CAM. <https://doi.org/10.13140/RG.2.2.11183.24484>
- [4]. Azubuike, C., & Obiefuna, O. (2014). Wireless communication: The impact of GSM on the economic lives of the Nigerian rural users. *Journal of Educational and Social Research*, 4(7), 79–86. <https://doi.org/10.5901/jesr.2014.v4n7p79>
- [5]. Chowdhury, A. M. M. I., & Hasan, E. S. (2025). Smart surveillance systems: Trends, challenges, and future directions. *Indonesian Journal of Computer Science*, 14(2), 1–10.
- [6]. Dutta, P., Garapati, U. K., Giribabu, Sridharan, B., & Agarwal, V. (2014). A comprehensive review of embedded system design aspects for rural application platform. *International Journal of Computer Applications*, 106(11), 39–44. <https://doi.org/10.5120/18567-9819>
- [7]. Edozie, E., Eze, V. H. U., Imaculate, M., & Janat, W. (2023). Design and implementation of a smart surveillance security system. *International Journal of Computer Engineering and Technology*, 14(1), 28–36.
- [8]. Gopinath, P. G., Harika, M., Chandu, K., Reddy, K. B., Chandrika, B. K., & Gurucharan, P. (2024). Development of GSM-based surveillance system in remote areas using solar power. *Journal of Emerging Technologies and Innovative Research*, 11(4), 1–10. <https://www.jetir.org/papers/JETIR2404690.pdf>
- [9]. Ikpenyi, O. E., Abumere, O. E., & Amusan, J. A. (2022). Construction of GSM based home security alert system using passive infrared sensor. *World Journal of Advanced Research and Reviews*, 14(2), 648–657. <https://doi.org/10.30574/wjarr.2022.14.2.0447>
- [10]. Kothandapani, V., Kodidela, P., & Gurram, P. (2021). IoT based smart intruder detection system for smart homes. *International Journal of Scientific Research in Science and Technology*. <https://doi.org/10.32628/IJSRST218410>



- [11]. Kristensen, F., Hedberg, H., Jiang, H., & Nilsson, P. (2008). An embedded real-time surveillance system: Implementation and evaluation. *Journal of Signal Processing Systems*, 52(1), 75–94. <https://doi.org/10.1007/s11265-007-0100-7>
- [12]. Likhitha, K., Malineni, S., Jampani, N., & Prasanna, N. L. (2019). Home security system using PIR sensor-IoT. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, 5(2), 497–500. <https://doi.org/10.32628/CSEIT195272>
- [13]. Mohapatra, B. N., Taware, S. S., & Parab, P. P. (2022). Smart security alarm system using PIR sensor. *Perspectives in Communication, Embedded-Systems and Signal-Processing (PiCES)*, 5(9), 90–92. <https://doi.org/10.5281/zenodo.5867913>
- [14]. Ohazuruike, K., Elechi, F. A., & Eze, P. N. (2024). Impact of herdsmen and community clashes on national integration in Nigeria. *South East Political Science Review*, 9(1). <https://journals.npsa-se.org.ng/index.php/SEPSR/article/view/211>
- [15]. Oladele, S. G., & Uk, S. I. O. (2018). Design and implementation of farm monitoring and security system. *International Journal of Computer Applications*, 182(31), 31–36.
- [16]. Pamadi, V. N., & Singh, P. (2025). Edge AI vs Cloud AI: A comparative study of performance, latency and scalability. *International Journal of Engineering and Emerging Technology*, 13(3).
- [17]. Rana, A., Kumar, T., & Puri, N. (2022, November). Design of GSM-based fire detection system using microcontroller and sensors. In *Advances in Manufacturing Technology and Management: Proceedings of the 6th International Conference on Advanced Production and Industrial Engineering (ICAPIE)* (pp. 557–568). Springer. https://doi.org/10.1007/978-981-16-9523-0_62
- [18]. Santamaria, A. F., Raimondo, P., Tropea, M., De Rango, F., & Aiello, C. (2019). An IoT surveillance system based on a decentralised architecture. *Sensors*, 19(6), 1462. <https://doi.org/10.3390/s19061462>
- [19]. Sá, P. H. M., Loureiro, R. B., Lisboa, F. V. N., & Peixoto, R. M. (2023, October). Efficient deployment of machine learning models on microcontrollers: A comparative study of quantization and pruning strategies. In *Proceedings of the IX Simpósio Internacional de Inovação e Tecnologia (SIINTEC)*. <https://doi.org/10.5151/siintec2023-305873>

